

Artificial Intelligence

Core Concepts, Myths vs. Facts, and the 8 Essential Terms Every Beginner Must Know — explained simply, with examples and diagrams.

AI · ML · DL · Gen AI

Myths vs Facts

Core Terminology

Diagrams & Examples

What this document covers

1. The AI Family — AI, Machine Learning, Deep Learning & Generative AI
2. AI: Myth vs. Fact (including why AI "hallucinates")
3. Why AI Matters
4. Why Terminology Matters
5. The 8 Core AI Terms — Algorithm, Training Data, Model, Training, Parameters, Inference, Token, Prompt
6. Types of Machine Learning — Predictive vs. Generative, Supervised, Unsupervised & Reinforcement Learning
7. Generative AI & LLM Foundations — Foundation, Multimodal & Unimodal Models, NLP, LM vs. LLM
8. How LLMs Are Built — The End-to-End Model Lifecycle
9. Retrieval-Augmented Generation (RAG) — Concept, Types & Pipeline
10. Quick Revision Summary Sheet

1 The AI Family — Core Definitions

Artificial Intelligence is not one single technology — it is an umbrella term that contains several nested fields, each narrower and more specialised than the one before it. Understanding this nesting is the fastest way to stop confusing the terms.

Artificial Intelligence (AI)

Definition: The broad field of building machines/software that can perform tasks which normally require human intelligence — such as reasoning, understanding language, recognising images, or making decisions.

Simple way to think about it: AI is the goal ("make a machine act smart"); everything else in this section is a technique used to reach that goal.

Example: A chess-playing computer, a voice assistant, and a self-driving car are all applications of AI.

Machine Learning (ML)

Definition: A subset of AI where, instead of being explicitly programmed with fixed rules, a system **learns patterns from data** and improves its performance with experience.

Example: An email system that learns to flag spam by studying thousands of past emails, rather than following a hand-written rulebook.

Deep Learning (DL)

Definition: A subset of Machine Learning that uses multi-layered **artificial neural networks** (loosely inspired by the human brain) to automatically learn complex patterns from large amounts of data.

Example: Face-unlock on a smartphone uses deep learning to recognise your face from many layers of learned visual features.

Generative AI (Gen AI)

Definition: A subset of Deep Learning focused on **creating new content** — text, images, audio, code, or video — rather than just classifying or predicting existing data.

Example: A chatbot writing an essay, or a tool generating an image from a text description.

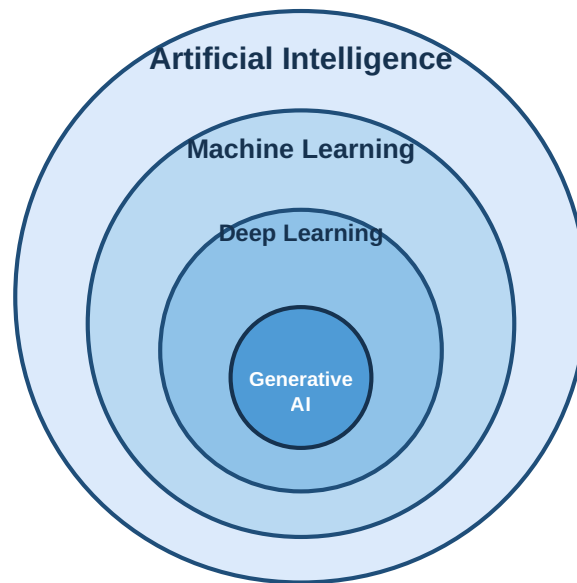


Fig. 1 — Each field is nested inside the broader one: AI $\supset$ Machine Learning $\supset$ Deep Learning $\supset$ Generative AI.

EXAMPLE — GPT-5 AND THE IDEA OF A "TRAINING PERIOD"

GPT-5 is a real-world example of a **Generative AI** model built using **Deep Learning**. Before it can be used, it goes through a **training period**, where it studies enormous amounts of text data to learn language patterns. Once training ends, the model is deployed and answers users' questions using **inference** (see Term 6) — it is no longer learning from those conversations in real time. This training-then-use pattern applies to almost all modern AI models.

2 AI: Myth vs. Fact

Because AI is widely discussed in media and marketing, several misconceptions have become common. Separating myth from fact is essential for using AI responsibly.

Myth	Fact
AI is always correct and objective.	AI can be confidently wrong. It reflects patterns and biases present in its training data and can produce false information (see "Hallucination" below).
AI understands language and meaning the way humans do.	AI predicts statistically likely patterns in data; it does not possess genuine comprehension, consciousness, or intent.
AI is a single, unified technology.	"AI" is an umbrella term covering many distinct techniques — ML, DL, Generative AI, rule-based systems, and more (see Section 1).
More data always means a smarter AI.	Quality and diversity of data matter as much as quantity — poor or biased data produces a poor or biased model (see Term 2: Training Data).

Spotlight: Why AI "Hallucinates"

KEY TERM — HALLUCINATION

A **hallucination** is when an AI model generates output that sounds fluent and confident but is factually incorrect, fabricated, or not grounded in real data.

Why it happens: Generative AI models work by predicting the most statistically likely next word/token based on patterns learned during training — they are not querying a verified database of facts. When the model lacks reliable information on a topic, it can still produce a plausible-sounding — but wrong — answer, because "sounding plausible" is what it was trained to do.

Practical takeaway: Always verify critical facts, figures, citations, or dates produced by an AI system against a trusted source.

3 Why AI Matters

AI is reshaping how work and daily life are done because it can process large amounts of data and perform certain tasks faster and at greater scale than humans. Its impact spans nearly every industry:

⚙️ Automation of repetitive tasks

📈 Faster, data-driven decisions

🏥 Improved healthcare diagnostics

🎯 Personalised recommendations

💬 Natural language interaction

🔍 Pattern detection at scale

KEY POINT

AI matters because it changes *how* problems are solved — shifting from rigid, manually-coded rules to systems that learn and adapt from data, enabling capabilities (like natural conversation or image generation) that were impractical to hand-code.

4 Why Terminology Matters

AI terminology is often used loosely or interchangeably in everyday conversation, which leads to confusion. Knowing the precise meaning of core terms allows you to:

- Understand how an AI system actually works "under the hood," rather than treating it as a black box.
- Communicate clearly with technical teams, articles, and documentation.
- Evaluate claims about AI products critically (e.g., knowing the difference between a model being *trained* versus performing *inference*).

The next section builds a precise, practical vocabulary of the 8 terms every AI learner should know.

5 The 8 Core AI Terms

Term 1 — Algorithm

Definition: A precise, step-by-step set of rules or instructions a computer follows to solve a problem or reach a decision.

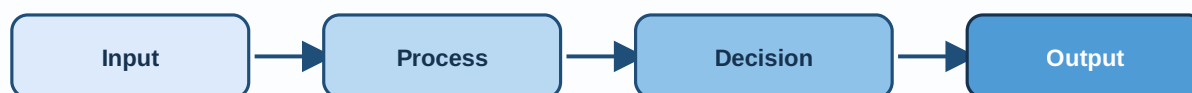


Fig. 2 — The general algorithm flow: Input → Process → Decision → Output.

Example: A spam-filter algorithm takes an email (input), scans it for suspicious words and sender details (process), decides whether it looks like spam (decision), and moves it to the Inbox or Spam folder (output).

Term 2 — Training Data

Definition: The dataset used to teach a machine learning model the patterns it needs to perform its task. The model is only as good as the data it learns from.

Dimension	What it means	Why it matters
Quality	Data is accurate, clean, and correctly labelled.	Poor-quality data teaches the model wrong patterns ("garbage in, garbage out").
Quantity	Enough examples exist for the model to learn general, reliable patterns.	Too little data leads to weak, unreliable predictions.
Diversity	Data covers a wide range of cases, styles, and scenarios.	Low diversity causes bias and poor performance on unseen or underrepresented cases.

Example: To train a model to recognise cats, it should see many (quantity) clear (quality) photos of cats of different breeds, colours, poses, and lighting conditions (diversity).

Term 3 — Model

Definition: The trained mathematical system that has learned patterns from training data and can be used to make predictions or generate outputs on new, unseen inputs. A model is the *result* of the training process.

Example: GPT-5 is a language model; a "spam vs. not-spam" classifier is also a model.

Common AI Task Types

Trained models are typically built for a specific category of task:

Task Type	What it does	Example
Language	Understanding or generating human text	Chatbots, translation, summarisation
Image	Recognising, classifying, or generating images	Face unlock, medical scan analysis
Speech	Converting speech to text or generating speech	Voice assistants, transcription apps
Forecasting	Predicting future values from historical trends	Sales forecasts, weather prediction
Recommendation	Suggesting relevant items to a user	Netflix, Amazon, Spotify suggestions

Term 4 — Training

Definition: The process of repeatedly feeding training data through an algorithm so the model can adjust its internal settings and gradually reduce errors until it reliably learns the underlying patterns.

Simple analogy: Like a student practising many past exam papers and correcting mistakes each time, until performance improves.

Term 5 — Parameters

Definition: The internal values (also called weights) inside a model that are automatically adjusted during training. Together, they represent everything the model has "learned."

Example: Large language models are often described by their parameter count (e.g., "billions of parameters") — broadly, more parameters can allow a model to capture more complex patterns, though quality of data and training matter just as much.

Term 6 — Inference

Definition: The process of using an already-trained model to generate a prediction or response for new input. This is what happens every time you actually *use* an AI system.

Key distinction: Training happens (typically) once, in advance, to build the model. Inference happens every single time a user interacts with it afterwards.

Term 7 — Token

Definition: A small unit of text — a whole word, part of a word, or a character — that a language model reads and generates internally. Text is broken into tokens before the model can process it.

Example: The word "unbelievable" might be split into tokens such as "un," "believ," "able."

Term 8 — Prompt

Definition: The instruction, question, or input text a user provides to an AI model to get a desired response.

Practical tip: Clear, specific, and well-structured prompts generally produce more accurate and useful outputs.

How the Terms Connect — End-to-End Flow

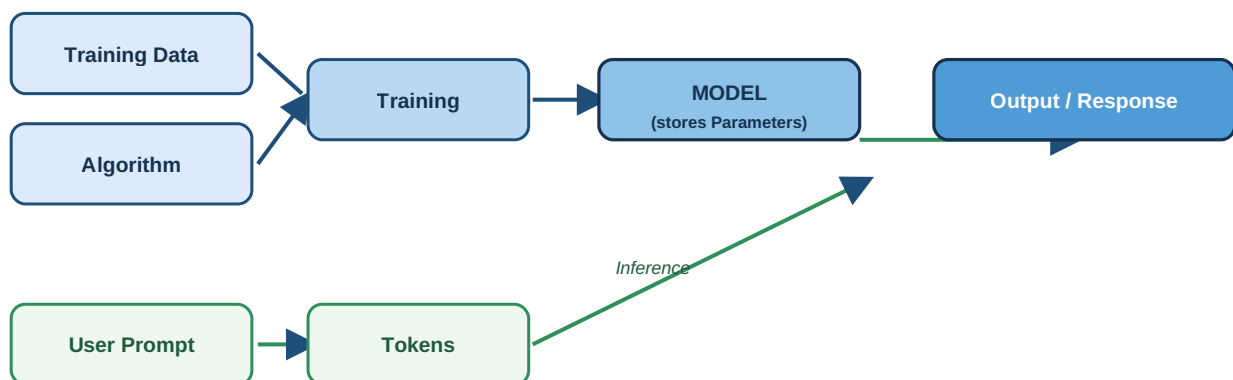


Fig. 3 — Training phase (top, blue) builds the Model; Inference phase (bottom, green) uses the Model with a new Prompt to produce a Response.

6 Types of Machine Learning

Machine Learning systems can be categorised in two complementary ways: by **what they are used for** (Predictive vs. Generative), and by **how they learn** (Supervised, Unsupervised, or Reinforcement Learning).

Category	What it does
Predictive AI	Analyses existing data to predict an outcome or classify new data into known categories. Focused on answering "what is this?" or "what will happen?"
Generative AI	Learns the underlying patterns of data well enough to create brand-new content (text, images, audio, code) rather than just predicting a label.

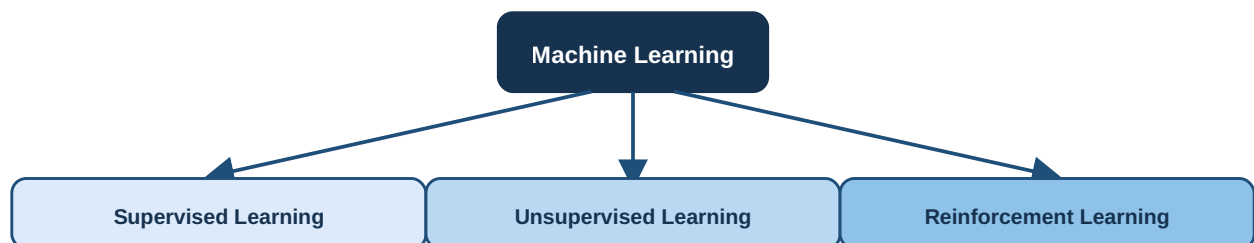


Fig. 4 — Machine Learning is commonly split into three learning styles.

Learning Style	How it learns	Sub-types	Example
Supervised	Trained on <i>labelled</i> data — each input has a known, correct output.	Classification (predicts a category) Regression (predicts a number)	Spam vs. not-spam (classification); predicting house prices (regression).
Unsupervised	Trained on <i>unlabelled</i> data — the model finds hidden structure on its own.	Clustering Association Dimensionality Reduction	Customer segmentation (clustering); "customers who buy X also buy Y" (association); compressing data for visualisation (dimensionality reduction).
Reinforcement	An agent takes actions in an environment and learns from reward/penalty feedback to maximise long-term reward.	Agent $\rightleftharpoons$ Environment loop	A game-playing AI that gets a reward for winning moves and a penalty for losing ones; robotics control.

KEY POINT

"Predictive vs. Generative" describes the *purpose* of a model; "Supervised / Unsupervised / Reinforcement" describes *how it learns*. A single Generative AI model, for instance, is typically trained using a mix of these learning styles at different stages (see Section 8).

7 Generative AI & LLM Foundations

Foundation Model

Definition: A large, general-purpose model pre-trained on broad, wide-ranging data, which can then be adapted (fine-tuned) to many different downstream tasks instead of being built from scratch each time.

Multimodal Model

Definition: A model that can understand and/or generate **multiple types of data** — for example text, images, and audio together.

Example: A model that can look at a photo and answer written questions about it.

Unimodal Model

Definition: A model that works with only **one type of data** — for example, a model that only processes text, or only processes images.

Natural Language Processing (NLP)

Definition: The field of AI concerned with teaching machines to **understand, interpret, and generate** human language.

LLM vs. LM

Aspect	Language Model (LM)	Large Language Model (LLM)
Scale	Smaller, with far fewer parameters.	Massive — typically billions of parameters.
Training data	Limited or domain-specific text.	Huge, diverse, internet-scale text corpora.
Capability	Basic next-word prediction; narrow tasks.	Broad language understanding, reasoning, and generation across many tasks.
Example	Small, task-specific text-prediction models.	GPT-5 and similar large-scale models.

Key point: Every LLM is a Language Model, but not every Language Model is "large" enough to be called an LLM.

8 How LLMs Are Built — The Model Lifecycle

Building a Large Language Model follows a broad, repeatable pipeline, from gathering raw data through to putting the finished model in users' hands.

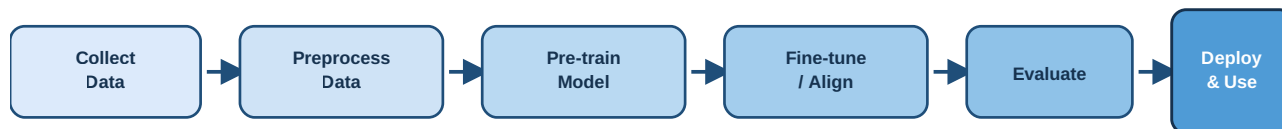


Fig. 5 — The end-to-end LLM lifecycle, from raw data to a deployed model.

Stage	What happens
Collect Data	Gather large, varied sources of text (and other data) to learn from.
Preprocess Data	Clean, filter, and tokenise the raw data so it is ready for training.
Pre-train Model	The model repeatedly practises predicting the next token across the entire dataset — this is the resource-intensive "training period" (see Section 1's GPT-5 example).
Fine-tune / Align	The pre-trained model is further trained on curated examples or human feedback so it becomes more accurate, safe, and helpful for real tasks.
Evaluate	The model's accuracy, safety, and reliability are tested before release.
Deploy & Use	The finished model is released for real users — every interaction from here on is Inference (see Term 6, Section 5).

9 Retrieval-Augmented Generation (RAG)

DEFINITION

RAG (Retrieval-Augmented Generation) is an AI framework that combines a **retrieval system** with a **generative model**: instead of relying only on what an LLM memorised during training, RAG fetches relevant external documents at the moment of the query and feeds them to the LLM so it can generate a grounded, up-to-date answer.

Why RAG Is Needed

- **Outdated knowledge:** An LLM's knowledge is frozen at its training cut-off (see the GPT-5 "training period" example in Section 1) — RAG lets it access current information.
- **Reduces hallucination:** Grounding answers in retrieved, real documents makes fabricated answers (see Section 2) less likely.

- **Access to private/custom data:** Organisations can connect an LLM to their own documents without expensive retraining.

Two Key Components

 **Retrieval** — find relevant information

 **Generation** — LLM composes the answer

Types of RAG

Type	What it means
Naive RAG (Basic)	The simplest form — retrieve relevant documents, then pass them straight to the LLM to generate an answer.
Advanced RAG	Adds optimisation steps before/after retrieval (e.g., query rewriting, re-ranking results) to improve the relevance of retrieved content.
Modular RAG	A flexible, modular pipeline where retrieval and generation components can be swapped or reconfigured for a specific use case.
Agentic RAG	An AI agent dynamically plans and orchestrates multiple retrieval and reasoning steps — sometimes using external tools — to answer complex, multi-part queries.

FRAMEWORK NOTE

LangChain is a widely used open-source AI framework for building RAG applications — it provides ready-made building blocks to connect LLMs, retrievers, and vector databases together.

Stage 1 — Ingestion Pipeline

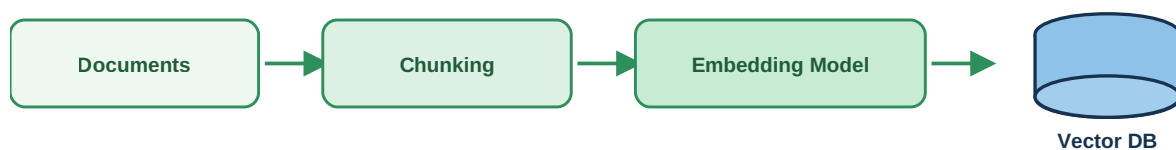


Fig. 6 — Ingestion pipeline: raw documents are chunked, embedded, and stored in a vector database.

Stage 2 — Retrieval & Generation Pipeline

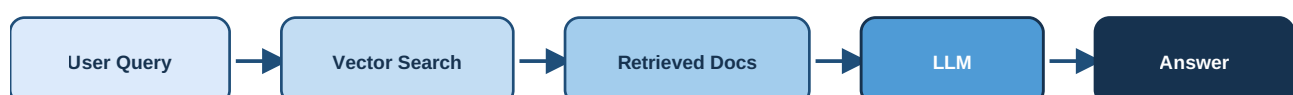


Fig. 7 — Retrieval & Generation pipeline: the user's query is used to search the vector database, and the retrieved context is passed to the LLM to generate the final answer.

PRACTICAL TIP — VIRTUAL ENVIRONMENTS

When building a RAG application with a framework like LangChain, it's good practice to work inside a Python **virtual environment**. To check which dependencies/packages are already installed in it, run:

- `pip list` — shows installed packages, or
- `pip freeze` — shows installed packages in a requirements-file format.

Quick Revision Summary Sheet

The AI Family

Term	One-line definition
AI	Machines performing tasks that need human-like intelligence.
ML	Systems that learn patterns from data instead of fixed rules.
DL	ML using multi-layered neural networks for complex patterns.
Gen AI	DL that creates new content (text, image, audio, code).

The 8 Core Terms

#	Term	One-line definition
1	Algorithm	Step-by-step rules: Input → Process → Decision → Output.
2	Training Data	Dataset used to teach the model (Quality, Quantity, Diversity).
3	Model	The trained system that makes predictions on new data.
4	Training	Process of learning from data by adjusting parameters.
5	Parameters	Internal learned values ("weights") inside a model.
6	Inference	Using a trained model to respond to new input.
7	Token	Small unit of text (word/sub-word) a model processes.
8	Prompt	The instruction/question a user gives to the AI.

Types of Machine Learning

Type	One-line definition
Predictive AI	Predicts outcomes or classifies data based on existing patterns.
Generative AI	Creates brand-new content rather than just predicting a label.
Supervised	Learns from labelled data (Classification, Regression).
Unsupervised	Finds hidden patterns in unlabelled data (Clustering, Association, Dimensionality Reduction).
Reinforcement	An agent learns via reward/penalty feedback from an environment.

Generative AI & LLM Foundations

Term	One-line definition
Foundation Model	Large, general-purpose model pre-trained for many downstream tasks.
Multimodal Model	Works with multiple data types (e.g., text + image).
Unimodal Model	Works with only one data type.
NLP	Teaching machines to understand, interpret & generate language.
LM vs. LLM	LM = any language-predicting model; LLM = a massive-scale LM (billions of parameters).

RAG at a Glance

Concept	One-line definition
RAG	Combines retrieval of external documents with LLM generation for grounded answers.
Naive RAG	Basic retrieve-then-generate pipeline.
Advanced RAG	Adds query/result optimisation around retrieval.
Modular RAG	Swappable, reconfigurable pipeline components.
Agentic RAG	An agent plans multi-step retrieval & reasoning dynamically.
Ingestion Pipeline	Documents → Chunking → Embedding Model → Vector DB.
Retrieval & Generation	User Query → Vector Search → Retrieved Docs → LLM → Answer.

REMEMBER

Training happens once to build the Model; Inference happens every time you use it. Hallucinations occur because generative models predict plausible patterns, not verified facts — RAG helps reduce this by grounding answers in retrieved, real documents. Always double-check important outputs.